

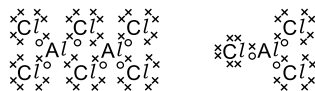
- 1 Proton, p , being positively charged will be attracted to the negative plate, while electron, e , being negatively charged will be attracted to the positive plate.

$$\text{Angle of deflection, } \theta \propto \frac{\text{charge, } z}{\text{mass, } m}$$

$$\left| \frac{z}{m}(e) \right| = \frac{1.60 \times 10^{-19}}{9.11 \times 10^{-31}} \gg \frac{1.60 \times 10^{-19}}{1.67 \times 10^{-27}} = \left| \frac{z}{m}(p) \right|$$

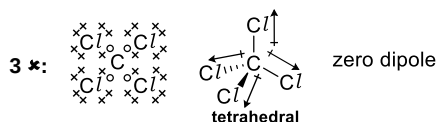
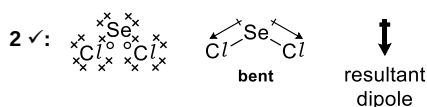
⇒ D

- 2 Gaseous Al_2Cl_6 and $AlCl_3$ are both simple molecular:



⇒ D

- 3 1 *: $\text{Se}=\text{C}=\text{Se}$ zero dipole
linear

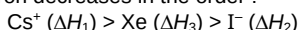


⇒ D

- 4 Cs^+ , I^- and Xe are *isoelectronic*, i.e. same number of electrons.

Since nuclear charge (number of protons) decreases in the order ${}_{55}\text{Cs}^+ > {}_{54}\text{Xe} > {}_{53}\text{I}^-$, while the shielding is identical, *effective nuclear charge* decreases in the order $\text{Cs}^+ > \text{Xe} > \text{I}^-$.

As the outermost electrons will be more strongly held the higher the effective nuclear charge, the amount of energy required (endothermic) to remove an electron decreases in the order:



⇒ B

- 5 A *: One mole of a compound is the amount that contains the same number of molecules as there are atoms in 12.000g of carbon-12.

B *: Relative isotopic mass of lithium-7 is
 $\frac{\text{mass of one atom of lithium-7}}{\frac{1}{12} \text{ the mass of one atom of carbon-12}}$

C ✓: Average mass of one atom of oxygen is calculated over the distribution of all the isotopes of oxygen.

D *: Relative molecular mass of E is
 $\frac{\text{average mass of one molecule of E}}{\frac{1}{12} \text{ the mass of one atom of carbon-12}}$

⇒ C

- 6 Atomic radius increases down a group and decreases across the period (for the main group elements).

K (Group 1) and Br (Group 17) are in period 4, while Rb (Group 1) and In (Group 13) are in period 5 of the periodic table. Hence, the two group 1 metals will be the larger atom within each period, while Rb in period 5 will be larger than K in period 4.

⇒ D

- 7 partial pressure of O_2 in air at sea level
 $= \frac{1}{5} \times 1 \text{ bar} = 0.200 \text{ bar}$

$$\begin{aligned} \text{percentage (by volume) of } \text{O}_2 \text{ in tank} &= \frac{\text{partial pressure of } \text{O}_2}{\text{total pressure in tank}} \times 100 \\ &= \frac{0.200 \text{ bar}}{4 \text{ bar}} \times 100 = 5\% \end{aligned}$$

⇒ C

- 8 A *: $\Delta H_{\text{lattice energy}}^\ominus (\text{Al}_2\text{O}_3)$ involves formation of **one mole of $\text{Al}_2\text{O}_3(\text{s})$** .

B *: $\Delta H_{\text{neutralisation}}^\ominus$ is the energy evolved when **one mole of $\text{H}_2\text{O}(\text{l})$** is formed from the reaction between an acid and a base.

C ✓: $\Delta H_{\text{solution}}^\ominus (\text{CaCl}_2(\text{s}))$ is the enthalpy change when **one mole of solid CaCl_2** is dissolved in water to give an infinitely diluted solution of the ions.

D *: $\Delta H_{\text{formation}}^\ominus (\text{SO}_2(\text{g}))$ is the enthalpy change when **one mole of gaseous SO_2** is formed from the constituent elements in their standard states.

⇒ C

- 9 The standard enthalpy change for
 $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \xrightarrow{\Delta H^\ominus} \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$
 can be obtained from the bond energies:

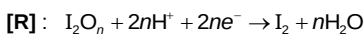
$$\begin{aligned} \Delta H_1^\ominus &= \sum \text{B.E. (bonds broken)} \\ &\quad - \sum \text{B.E. (bonds formed)} \\ &= 4 \times \text{B.E. (C-H)} + 2 \times \text{B.E. (O=O)} \\ &\quad - [2 \times \text{B.E. (C=O)} + 4 \times \text{B.E. (O-H)}] \\ &= 4 \times 410 + 2 \times 496 - (2 \times 805 + 4 \times 460) \\ &= -818 \text{ kJ mol}^{-1} \end{aligned}$$

Hence, the standard enthalpy of vapourisation of water, $\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{g})$:

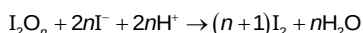
$$\begin{aligned} \Delta H_{\text{vapourisation}}^\ominus (\text{H}_2\text{O}(\text{l})) &= \frac{\Delta H_1^\ominus - \Delta H_c^\ominus (\text{CH}_4(\text{g}))}{2} \\ &= \frac{-818 - (-890)}{2} \\ &= +36 \text{ kJ mol}^{-1} \end{aligned}$$

⇒ C

- 10 Let oxidation state of I in the oxide be n . So the oxide is I_2O_n .



Combining the two half-equations,



Since 0.001 mole of the oxide reacts with 0.01 mole of I^- to give 0.006 mole of I_2 , so 1 mole of the oxide will react with 10 moles of I^- give 6 moles of I_2 .

$$\text{Hence, } 2n = 10 \text{ or } n + 1 = 6 \Rightarrow n = 5$$

⇒ B

- 11 Since the rate equation is $\text{rate} = k[\text{NO}_2]^2$, the slow step is **bimolecular**, involving **two molecules of NO_2 reacting** to give an intermediate, which will be consumed in the subsequent steps.

In addition, the two steps must **add up to give the overall stoichiometric equation**.

⇒ C

- 12 Given that $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \frac{1}{2}\text{O}_2$ is catalysed by the **enzyme** catalase.

Enzyme-catalysed reactions display **saturation kinetics**.

For a given enzyme concentration and for relatively low $[\text{H}_2\text{O}_2]$, initial rate increases **linearly** with $[\text{H}_2\text{O}_2]$; the enzyme molecules are largely free to catalyse the reaction.

However, at relatively high $[\text{H}_2\text{O}_2]$, initial rate **asymptotically approaches the theoretical maximum**; the enzyme active sites are almost all occupied by H_2O_2 molecules resulting in saturation.

⇒ D

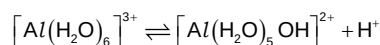
- 13 $r_{\text{Al}^{3+}} = 0.050 \text{ nm}$; $r_{\text{Cu}^{2+}} = 0.073 \text{ nm}$;

$$r_{\text{Mg}^{2+}} = 0.065 \text{ nm}; r_{\text{Na}^+} = 0.095 \text{ nm}$$

Hence, Al^{3+} has the highest charge

density $\left(\propto \frac{\text{charge}}{\text{ionic radius}} \right)$ and polarising

power, thus further polarises the O-H in the coordinated H_2O molecules to a greater extent, leading to greater ease of deprotonation to give $\text{H}^+(\text{aq})$:



⇒ A

- 14 In pure water, $[\text{H}^+] = [\text{OH}^-]$ as one H^+ and one OH^- will be formed when the O-H bond in one H_2O molecule cleaves.

$$\begin{aligned} \text{At } 10^\circ\text{C}, [\text{H}^+] &= \sqrt{0.293 \times 10^{-14}} \\ &= 5.413 \times 10^{-8} \text{ mol dm}^{-3} \end{aligned}$$

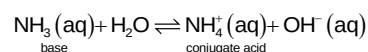
$$\text{pH} = -\lg[\text{H}^+] = 7.27$$

$$\begin{aligned} \text{At } 40^\circ\text{C}, [\text{H}^+] &= \sqrt{2.92 \times 10^{-14}} \\ &= 1.709 \times 10^{-7} \text{ mol dm}^{-3} \end{aligned}$$

$$\text{pH} = -\lg[\text{H}^+] = 6.77$$

⇒ A

- 15 A basic buffer is required to maintain the pH at about 10, comprising of a **weak base and its conjugated acid**:



⇒ A

- 16 1 ✓: $(\Delta G_1 + \Delta G_2)$ is ΔG for dissolving of $\text{AgCl}(\text{s})$ in $\text{NH}_3(\text{aq})$; $(\Delta G_3 + \Delta G_4)$ is ΔG for dissolving of $\text{AgBr}(\text{s})$ in $\text{NH}_3(\text{aq})$. As **solid AgCl** is more soluble than **solid AgBr** in $\text{NH}_3(\text{aq})$, the former must be more exergonic, i.e. ΔG is more negative. Thus, $(\Delta G_1 + \Delta G_2) < (\Delta G_3 + \Delta G_4)$.

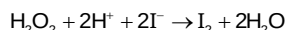
2 ✓: Eqm 2 and 4 are **identical**, involving the formation of $[\text{Ag}(\text{NH}_3)_2]^+(\text{aq})$ from $\text{Ag}^+(\text{aq})$ and $\text{NH}_3(\text{aq})$, with Cl^- and Br^- being spectator ions. Hence, $\Delta G_2 = \Delta G_4$.

3 *: From 2, $\Delta G_2 = \Delta G_4$.

4 ✓: Since **solid AgCl** is more soluble than **solid AgBr** in water, ΔG_1 must be more exergonic, i.e. more negative, than ΔG_3 . Thus $\Delta G_1 < \Delta G_3$.

⇒ A

- 17 I^- will reduce H_2O_2 to H_2O :

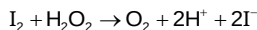


$$E_{\text{cell}}^{\ominus} = E^{\ominus}(H_2O_2|H_2O) - E^{\ominus}(I_2|I^-) \\ = +1.77 - (+0.54) = +1.23V > 0$$

Hence, the solution will turn **brown** due to the I_3^- formed: $I_2 + I^- \rightleftharpoons I_3^-$ (brown)

(I_2 is purple only when dissolved in a non-polar organic solvent like hexane)

However, the I_2 formed will **not** be able to oxidise the limited amount of H_2O_2 to O_2 :

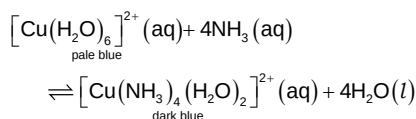


$$E_{\text{cell}}^{\ominus} = E^{\ominus}(I_2|I^-) - E^{\ominus}(O_2|H_2O_2) \\ = +0.54 - (+0.68) = -0.14V < 0$$

(Note: If a few drops of KI are added to excess H_2O_2 , I^- instead catalyses the decomposition of H_2O_2 to O_2 and H_2O)

⇒ B

- 18 The dark blue water-soluble complex ion formed between $Cu^{2+}(aq)$ and NH_3 is:

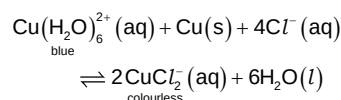


$Cu(OH)_2(H_2O)_4$ is a pale blue precipitate formed when OH^- is added to $Cu^{2+}(aq)$.

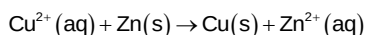
Copper(I) chloride, $CuCl$, is an insoluble white solid.

⇒ A

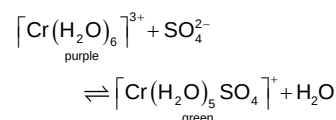
- 19 A ✖: Comproportionation of Cu and Cu(II) to give Cu(I):



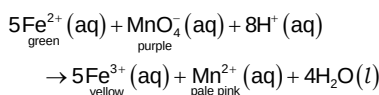
B ✖: This is a redox displacement reaction of Cu^{2+} by Zn:



C ✓: One of the sulfate ion functions as a monodentate ligand to displace a water ligand:



D ✖: Redox reaction:



⇒ C

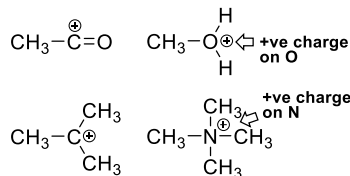
- 20 1 ✖: Due to diffuse shape of 3d orbitals, the outer 4s electrons in V are less shielded, hence experience greater effective nuclear charge (ENC), leading to smaller atomic radius compared to Ca, thus higher density.

2 ✖: Electrons in the outer shell do not shield one another effectively, hence the greater number of outer shell electrons does not explain for the higher ENC experienced.

3 ✓: Due to greater ENC (higher nuclear charge and poor shielding by the 3d electrons), the outer shell 4s electrons are more strongly attracted by the nucleus in V, accounting for the smaller atomic radius and hence higher density.

⇒ D

- 21 A carbocation is a positive ion with a significant portion of the excess positive charge located on one or more carbon atoms:

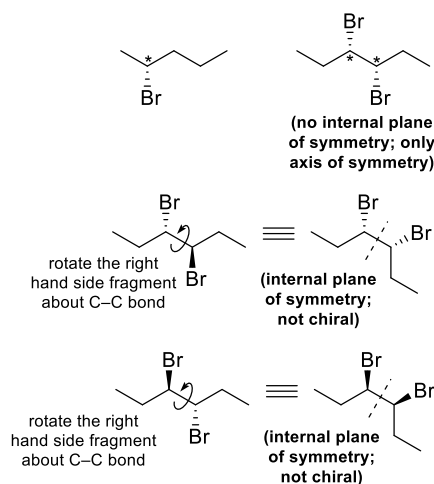


⇒ C

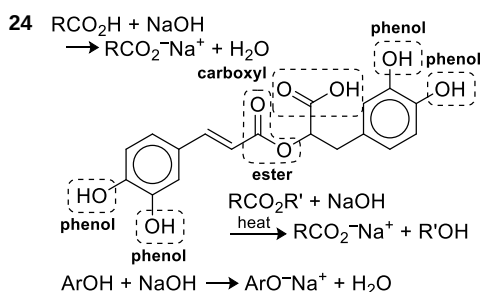
- 22 Tollen's reagent oxidises the **aldehyde** functional group, $\text{—}\overset{H}{\underset{O}{\parallel}}\text{—}$, to the carboxyl group.

⇒ A

- 23 Only chiral molecules are able to rotate plane polarised light.



⇒ B



⇒ D

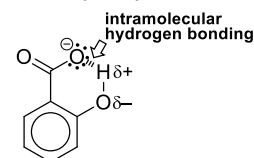
- 25 1 ✓: Due to the attack of the nucleophile from the rear side, with simultaneous expulsion of the leaving group, S_N2 occurs with **inversion of configuration** about the α -carbon.

2 ✓: Due to formation of a trigonal planar carbocation intermediate in the first fast step, the nucleophile can attack the carbocation from either face with equal probability, hence leading to a 1:1 mixture of the two enantiomers, hence **racemisation**.

3 ✓: 3° chloroalkanes refers to react via the S_N1 mechanism due to the **electron-donating effect** of the three alkyl groups which renders the 3° carbocation intermediate more stable. Also, the presence of three bulky alkyl groups impart steric hindrance, disfavours S_N2 attack.

⇒ A

- 26 The more stable the conjugate base, the more acidic is the organic acid. Intramolecular hydrogen bonding is present only in 2-hydroxybenzoate due to proximity of the OH and COO^- group, helping to stabilise the 2-hydroxybenzoate ion:

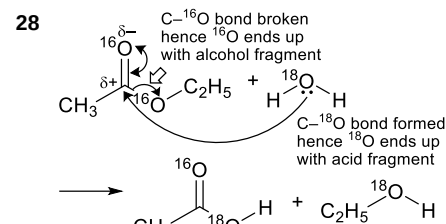


⇒ B

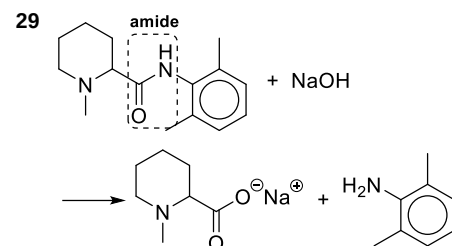
- 27 $H_2 + Ni$ only reduces the $C=C$ to saturated CH_2-CH_2 , but not the CO_2H

$LiAlH_4$ only reduces the CO_2H to a 1° alcohol, CH_2OH , but not the $C=C$ (since H^- is nucleophilic, so is the electron-rich $C=C$)

⇒ B



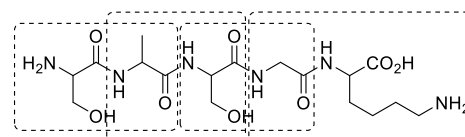
⇒ C



Only the amide group is hydrolysed to give a carboxylate salt and amine.

⇒ D

- 30 Only pentapeptide **D** allows all four dipeptides to be obtained:



⇒ D

Answer Key

Qn	Ans	Qn	Ans	Qn	Ans
1	D	11	C	21	C
2	D	12	D	22	A
3	D	13	A	23	B
4	B	14	A	24	D
5	C	15	A	25	A
6	D	16	A	26	B
7	C	17	B	27	B
8	C	18	A	28	C
9	C	19	C	29	D
10	B	20	D	30	D